

EC400: Solutions to Exam

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1. Denote the Lagrange multiplier with λ_t and write

$$\mathcal{L} = \sum_{t=1}^{\infty} \left\{ \beta^t \left[(p_t - \bar{p})^2 + \kappa \pi_t^2 \right] + \lambda_t (p_t - p_{t-1} - \pi_t) \right\}.$$

The first-order conditions wrt p_t and π_t are

$$\begin{aligned} 2\beta^t(p_t - \bar{p}) + \lambda_t &= \lambda_{t+1}, \\ 2\beta^t \kappa \pi_t &= \lambda_t. \end{aligned}$$

2. Combine together the previous equations to obtain

$$(p_t - \bar{p}) + \kappa \pi_t = \beta \kappa \pi_{t+1},$$

The transversality condition requires that

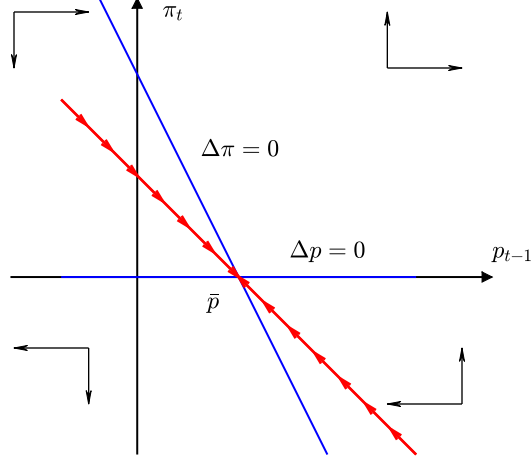
$$\lim_{t \rightarrow \infty} \lambda_t p_t = 0.$$

3. Notice that the dynamic system can be written as follows:

$$\begin{pmatrix} \beta\kappa & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \pi_{t+1} \\ p_t \end{pmatrix} = \begin{pmatrix} \kappa & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \pi_t \\ p_{t-1} \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \bar{p},$$

or equivalently,

$$\begin{pmatrix} \pi_{t+1} \\ p_t \end{pmatrix} = \begin{pmatrix} \frac{1+\kappa}{\beta\kappa} & \frac{1}{\beta\kappa} \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \pi_t \\ p_{t-1} \end{pmatrix} + \begin{pmatrix} -\frac{1}{\beta\kappa} \\ 0 \end{pmatrix} \bar{p}.$$



It follows that

$$\begin{aligned}\Delta\pi_{t+1} &= \frac{1}{\beta\kappa}(p_{t-1} - \bar{p}) + \left(\frac{1+\kappa}{\beta\kappa} - 1\right)\pi_t, \\ \Delta p_t &= \pi_t,\end{aligned}$$

and hence, the two isolines correspond to

$$\begin{aligned}\pi_t &= -\frac{1}{1+\kappa-\beta\kappa}(p_{t-1} - \bar{p}), \\ \pi_t &= 0,\end{aligned}$$

Imposing $\Delta p_t = \Delta\pi_{t+1} = 0$, we get the unique steady state $\pi = 0$, $p = \bar{p}$.¹

4. A (lagged) price level p_{t-1} is the state variable and inflation π_t (or equivalently, the present price level p_t) is the control variable. The phase diagram shows that the steady state is a saddle point. This is also in line with the Blanchard-Kahn principle: given one state variable and one control variable, a unique solution requires one eigenvalue greater than one and one eigenvalue less than one (in absolute terms).
5. Let p denote the state variable, i.e. the price level in the previous period p_{t-1} . The Bellman equation can then be written as

$$V(p) = \min_{\pi} \left\{ (p + \pi - \bar{p})^2 + \kappa\pi^2 + \beta V(p + \pi) \right\}.$$

¹Alternatively, one can derive the first isoline from $0 = \Delta\pi_t = \frac{1}{\beta\kappa}(p_{t-1} - \bar{p}) + \frac{1-\beta}{\beta}\pi_{t-1}$. Two answers coincide in the continuous-time limit, which implicitly assumed when drawing phase diagrams.

6. Substitute the guess into the Bellman equation

$$V(p) = \min_{\pi} \left\{ (p + \pi - \bar{p})^2 + \kappa\pi^2 + \beta\alpha(p + \pi - \bar{p})^2 \right\}$$

and take the first-order condition

$$p + \pi - \bar{p} + \kappa\pi + \beta\alpha(p + \pi - \bar{p}) = 0.$$

This implies that the policy function is given by

$$\pi(p) = -\frac{1 + \beta\alpha}{1 + \kappa + \beta\alpha}(p - \bar{p}).$$

Substitute this expression back into the Bellman equation

$$V(p) = (1 + \beta\alpha) \left(1 - \frac{1 + \beta\alpha}{1 + \kappa + \beta\alpha} \right)^2 (p - \bar{p})^2 + \kappa \left(\frac{1 + \beta\alpha}{1 + \kappa + \beta\alpha} \right)^2 (p - \bar{p})^2 = \frac{\kappa(1 + \beta\alpha)}{1 + \kappa + \beta\alpha} (p - \bar{p})^2.$$

It follows that

$$\alpha = \frac{\kappa(1 + \beta\alpha)}{1 + \kappa + \beta\alpha}.$$

Solve this quadratic equation

$$\beta\alpha^2 + (1 + (1 - \beta)\kappa)\alpha - \kappa = 0$$

to obtain two roots

$$\alpha_{1,2} = \frac{-(1 + (1 - \beta)\kappa) \pm \sqrt{(1 + (1 - \beta)\kappa)^2 + 4\beta\kappa}}{2\beta},$$

where $\alpha_1 > 0, \alpha_2 < 0$. Only the former root is consistent with the fact that the losses are increasing as the price gets further away from the desired level $\bar{p}$.

7. The price level is a stock measured *at* a particular point in time, while inflation is a flow measured *over* a period of time. Therefore, the law of motion can be rewritten for a period of length Δ as

$$p_t = p_{t-\Delta} + \pi_t \Delta.$$

Equivalently, we get

$$\frac{p_t - p_{t-\Delta}}{\Delta} = \pi_t$$

and taking the continuous-time limit $\Delta \rightarrow 0$,

$$\dot{p}_t = \pi_t.$$

8. Denote the co-state with λ_t and write down the Hamiltonian:

$$\mathcal{H}_t = e^{-\rho t} \left[(p_t - \bar{p})^2 + \kappa \pi_t^2 \right] + \lambda_t \pi_t.$$

Use the maximum principle:

$$\begin{aligned} 0 &= \frac{\partial \mathcal{H}_t}{\partial \pi_t} = 2e^{-\rho t} \kappa \pi_t + \lambda_t, \\ -\dot{\lambda}_t &= \frac{\partial \mathcal{H}_t}{\partial p_t} = 2e^{-\rho t} (p_t - \bar{p}), \\ 0 &= \lim_{t \rightarrow \infty} \lambda_t p_t. \end{aligned}$$

9. If $\pi_t = \gamma(p_t - \bar{p})$, then the first optimality condition implies that

$$\lambda_t = -2e^{-\rho t} \kappa \gamma (p_t - \bar{p}).$$

Using this expression, the second optimality condition can be expressed as

$$2\kappa\gamma \left[-\rho e^{-\rho t} (p_t - \bar{p}) + e^{-\rho t} \dot{p}_t \right] = 2e^{-\rho t} (p_t - \bar{p})$$

and simplified to

$$\pi_t = \dot{p}_t = \frac{1 + \kappa\gamma\rho}{\kappa\gamma} (p_t - \bar{p}).$$

This confirms the initial guess and implies that

$$\gamma = \frac{1 + \kappa\gamma\rho}{\kappa\gamma}.$$

This quadratic equation

$$\gamma^2 - \rho\gamma - \frac{1}{\kappa} = 0$$

has two roots

$$\gamma_{1,2} = \frac{\rho \pm \sqrt{\rho^2 + 4/\kappa}}{2}$$

with $\gamma_1 > \rho$ and $\gamma_2 < 0$. However, only the latter root satisfies the transversality

condition. Indeed, the differential equation for prices

$$\frac{dp_t}{p_t - \bar{p}} = \gamma dt$$

implies that

$$\log \frac{p_t - \bar{p}}{p_0 - \bar{p}} = \gamma t$$

or equivalently,

$$p_t - \bar{p} = e^{\gamma t}(p_0 - \bar{p}).$$

It follows that

$$\lambda_t p_t = -2\kappa\gamma e^{-\rho t}(p_t - \bar{p})p_t = -2\kappa\gamma e^{(\gamma-\rho)t}(p_0 - \bar{p})\left[\bar{p} + e^{\gamma t}(p_0 - \bar{p})\right].$$

Thus, the transversality condition $\lim_{t \rightarrow \infty} \lambda_t p_t = 0$ is not satisfied for $\gamma_1 > \rho$, but holds for $\gamma_2 < 0$.